Ziang Ni

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Delft, the Netherlands

Research interests

Reliability and factuality of LLM agents under social pressure; interpretability; multi-agent systems.

Education

Delft University of Technology Sep 2025 – Jul 2028 (expected)
B.Sc. in Computer Science and Engineering, GPA 8.0/10, on track for *cum laude*
Key courses: Calculus 9.5, Linear Algebra 9.0, Algorithms & Data Structures 9.0

Beijing Normal University Sep 2024 – Jun 2025
B.S. program in Artificial Intelligence; first year, then moved to TU Delft

Preprints

Silent Dissent: LLM Agents That Yield to the Majority Still Represent Their Original Premise
Ziang Ni*, Peng Zou* arXiv · Code *Preprint*

Measured or Real? When Model-Derived Measurements Can Be Corrected or Compared
Yifeng Li*, **Ziang Ni*** arXiv · Code *Preprint*

Research experience

Hidden beliefs of conforming LLM agents Sep 2026 – present
Independent research with Peng Zou

- Asked whether LLM agents that yield to a unanimous majority change their mind or only their answer, using an Asch-style debate on two-hop factual questions whose intermediate "bridge" entity is never stated.
- Reproduced Anthropic's Jacobian lens (Global Workspace, 2026) to read the bridge from the residual stream at the moment of answering.
- In pre-registered tests on held-out facts across four open-weight models, agents that gave in still represented their original bridge in three models (hit@100 above a control entity: 0.22–0.85), while the logit lens read almost none of it (0.00–0.06).
- In exploratory steering, injecting the bridge's J-lens direction restored the original answer in the two Qwen models; the negative results of the pre-registered program are reported alongside.

Reliability of stochastically trained models Jul 2026 – present
With Yifeng Li

- Co-developed a framework for how reliable scalar quantities from stochastic ML pipelines are (difficulty scores, representation measurements, prompt-sensitive leaderboard gaps), accounting for variation across seeds, replicate populations, and prompts.
- Developed a confidence-gated disattenuation procedure with exact-F reliability bounds and conservative finite-sample guarantees.

Markov logic networks for auditable LLM reasoning May 2026 – present
TU Delft, supervised by Prof. Jie Yang

- Built a Python framework that combines LLM-derived atom-level confidence scores with soft and hard logical constraints through explicit possible-world scoring.
- Applied it to a Board-of-Examiners case study, testing whether symbolic factors make LLM-assisted decisions in high-stakes administrative settings more transparent and auditable.

Other

The Last Math Competition Code Oct 2026 – present
#1 on the solver leaderboard
As of 19 Oct 2026, resolved 193 conjectures (16 proofs, 177 disproofs), all first solves, each with a LaTeX write-up and a Lean 4 formalization that passed full audit.

Optiver Trading Hackathon, Rotterdam 2025
Built an automated options-trading algorithm with Black–Scholes pricing and Greeks.

Skills

Interpretability and evaluation: logit lens, Jacobian lens, activation steering, pre-registered experimental design, controlled ablations, uncertainty quantification
Tools: Python, PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, C/C++, Bash, Linux, Git, LaTeX
Languages: Chinese (native), English (IELTS 7.5)

*Equal contribution